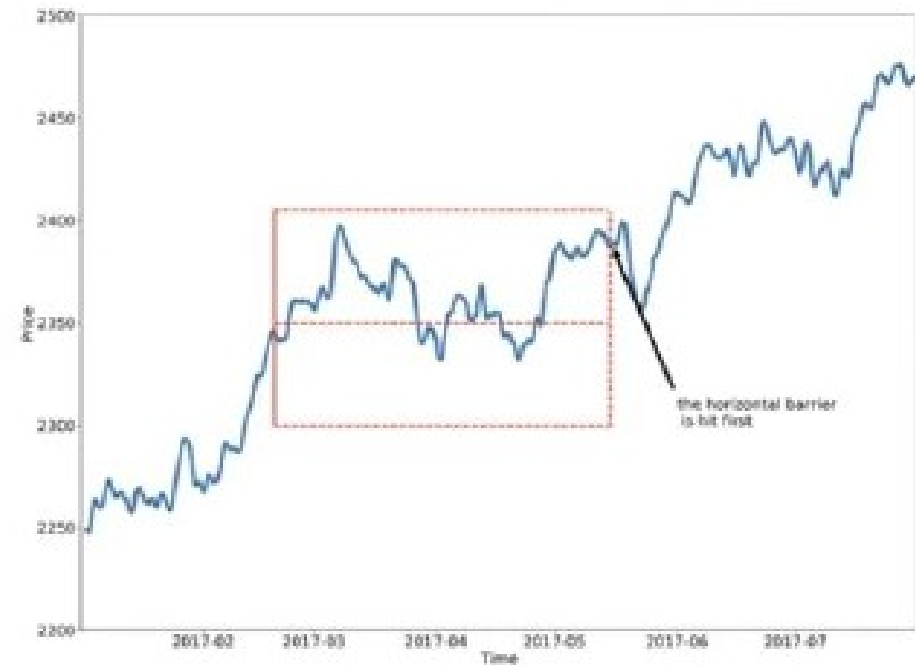
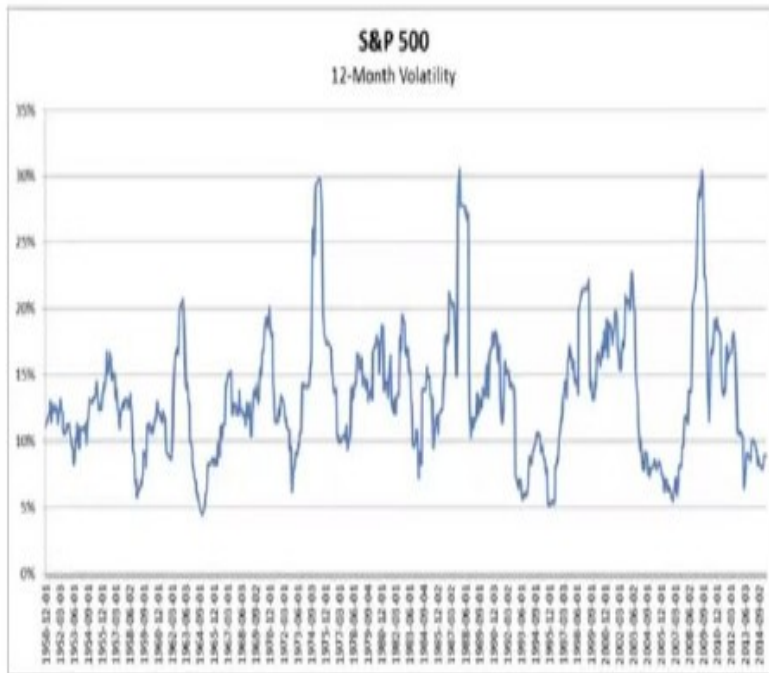


메타 레이블

삼중 배리어 레이블

□ 삼중 배리어 레이블 (triple barrier label)

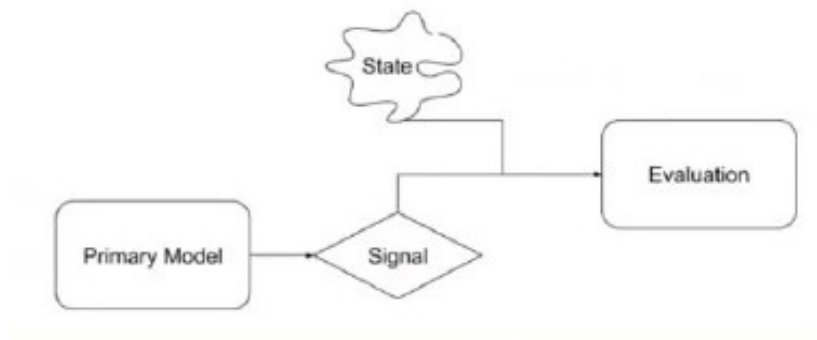


Triple-Barrier Label $y=0$ | Source: quantresearch.org

메타 레이블

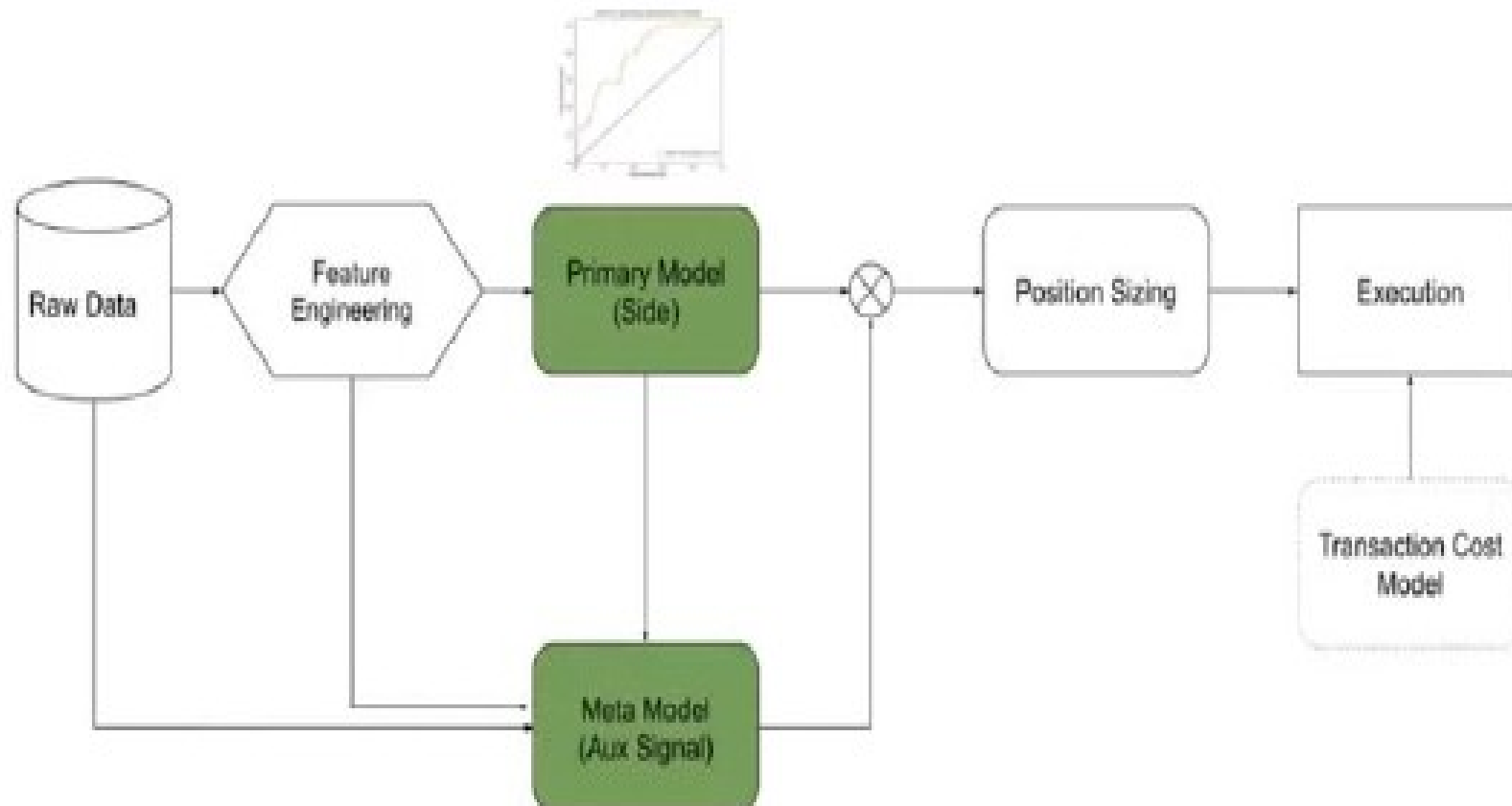
□ 트레이드 할 것인가 말 것인가?

- 베팅 사이드를 결정하는 모델을 가지고 있다. 이제 트레이드를 할 것인지를 결정해야 한다.
 1. 1차모델에서 사이드를 취한다. $[-1, 1]$
 2. 2차모델에서는 이것이 맞았는지 틀렸는지를 결정한다. 이를 통해 트레이드 결정을 한다.
- 1차모델은 재정거래, 기술적 투자기법, 전통적 퀀트, ML 모델 그 어떠한 것도 가능하다.
- 메타 레이블을 통해 재현율과 정밀도의 트레이드 오프를 개선하고자 한다. (즉 더 올바른 트레이드들 원한다.)



메타 레이블

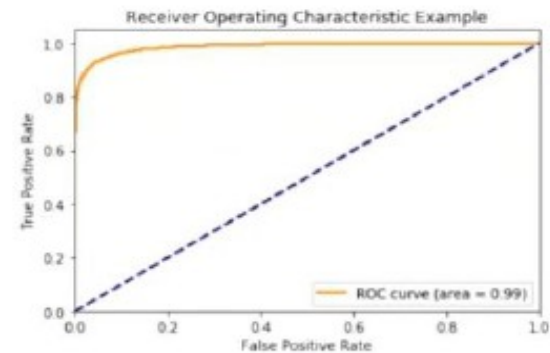
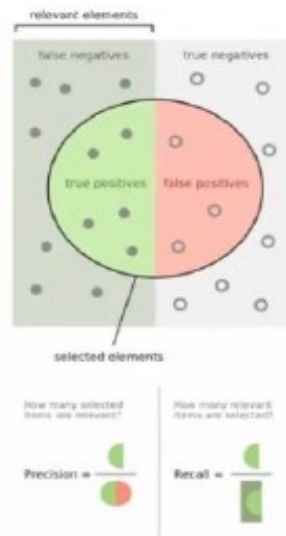
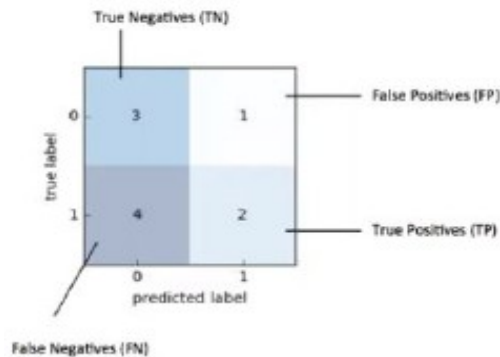
□ 트레이드 할 것인가 말 것인가?



메타 레이블의 원리

□ 메타 레이블의 작동원리

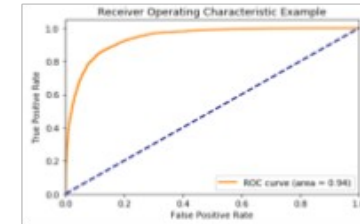
- 1차모델에서 높은 재현율을 원한다. (정밀도가 낮아도)
- 2차모델에서 1차모델에서 예측한 Positive(거래를 하자)에 메타 레이블을 적용해 낮은 정밀도를 개선한다.



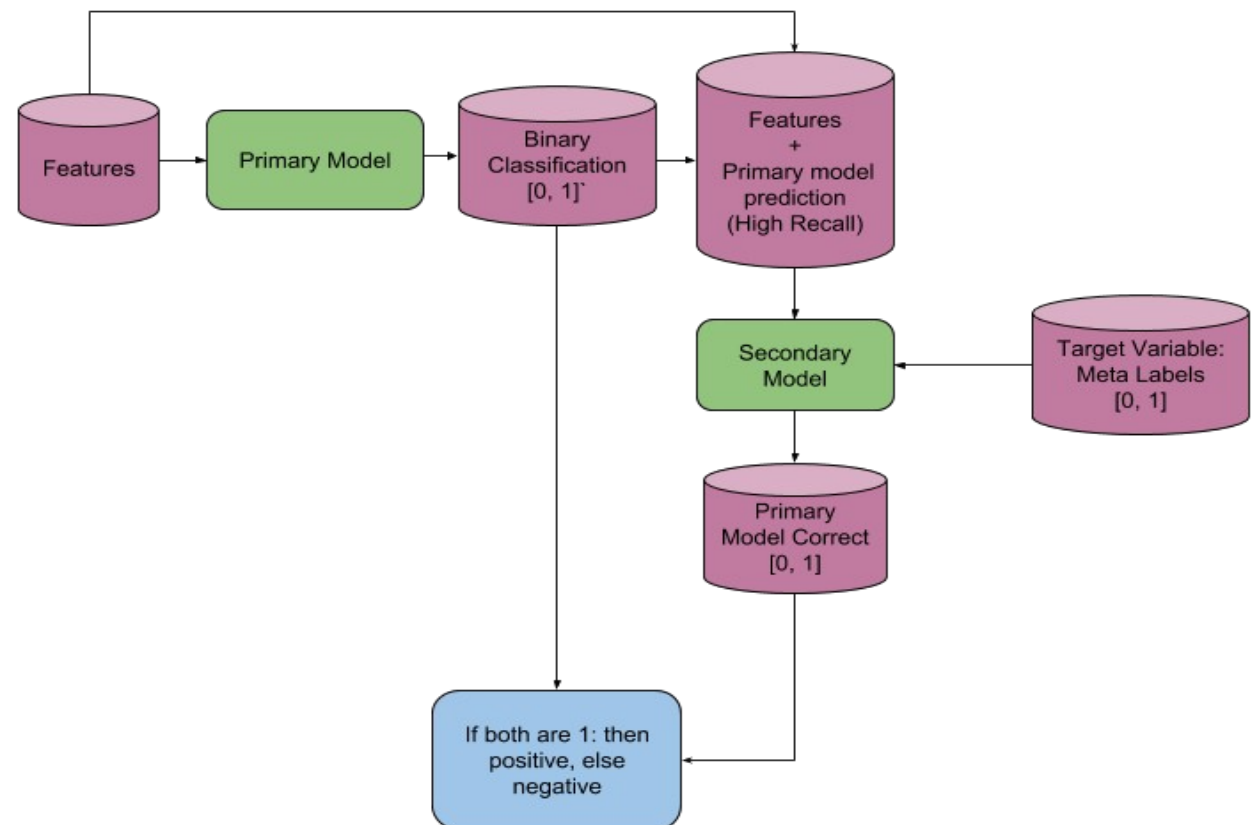
메타 레이블

□ 메타 레이블 구조 예제

Meta Labeling

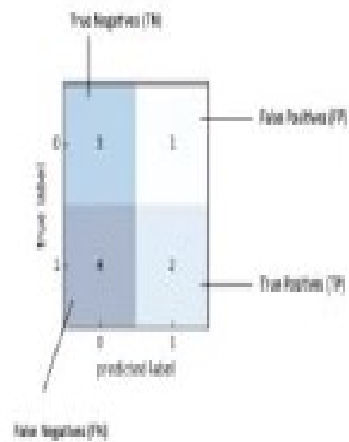


→ 5 3 5 →



메타 레이블

□ 메타 레이블 구조 예제



Base Model Metrics:

	precision	recall	f1-score	support
False	0.95	0.94	0.94	892
True	0.94	0.96	0.95	1010
micro avg	0.95	0.95	0.95	1902
macro avg	0.95	0.95	0.95	1902
weighted avg	0.95	0.95	0.95	1902

Confusion Matrix

```
[[700 192]
```

```
 [ 11 999]]
```

Accuracy: 0.8933

Meta Label Metrics:

	precision	recall	f1-score	support
False	0.95	0.96	0.95	892
True	0.96	0.95	0.96	1010
micro avg	0.96	0.96	0.96	1902
macro avg	0.96	0.96	0.96	1902
weighted avg	0.96	0.96	0.96	1902

Confusion Matrix

```
[[857  35]
```

```
 [ 47 963]]
```

Accuracy: 0.9569

메타 레이블

□ 메타 레이블 트레이딩 예제

	precision	recall	f1-score	support
0	0.00	0.00	0.00	749
1	0.17	1.00	0.29	151
micro avg	0.17	0.17	0.17	900
macro avg	0.08	0.50	0.14	900
weighted avg	0.03	0.17	0.05	900

Confusion Matrix
[[0 749]
[0 151]]

Accuracy
0.16777777777777778

	precision	recall	f1-score	support
0	0.85	0.68	0.75	749
1	0.20	0.41	0.27	151
micro avg	0.63	0.63	0.63	900
macro avg	0.53	0.54	0.51	900
weighted avg	0.74	0.63	0.67	900

Confusion Matrix
[[586 243]
[89 62]]

Accuracy
0.6311111111111111

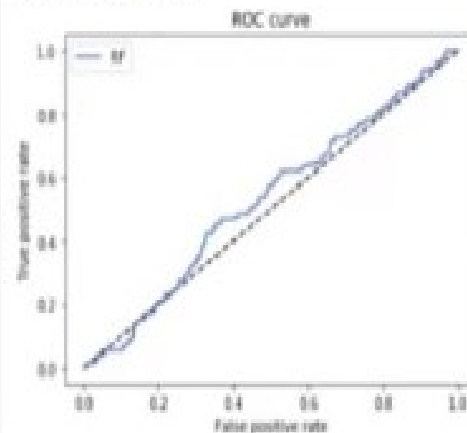
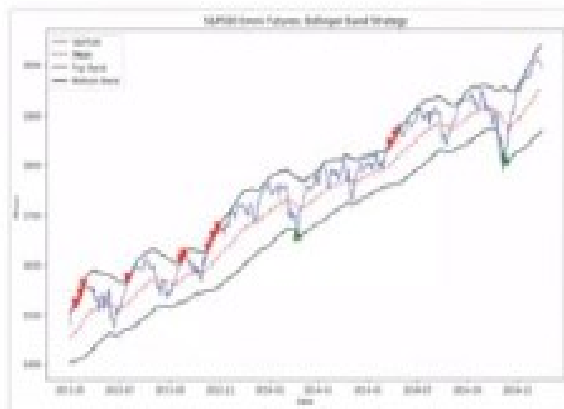
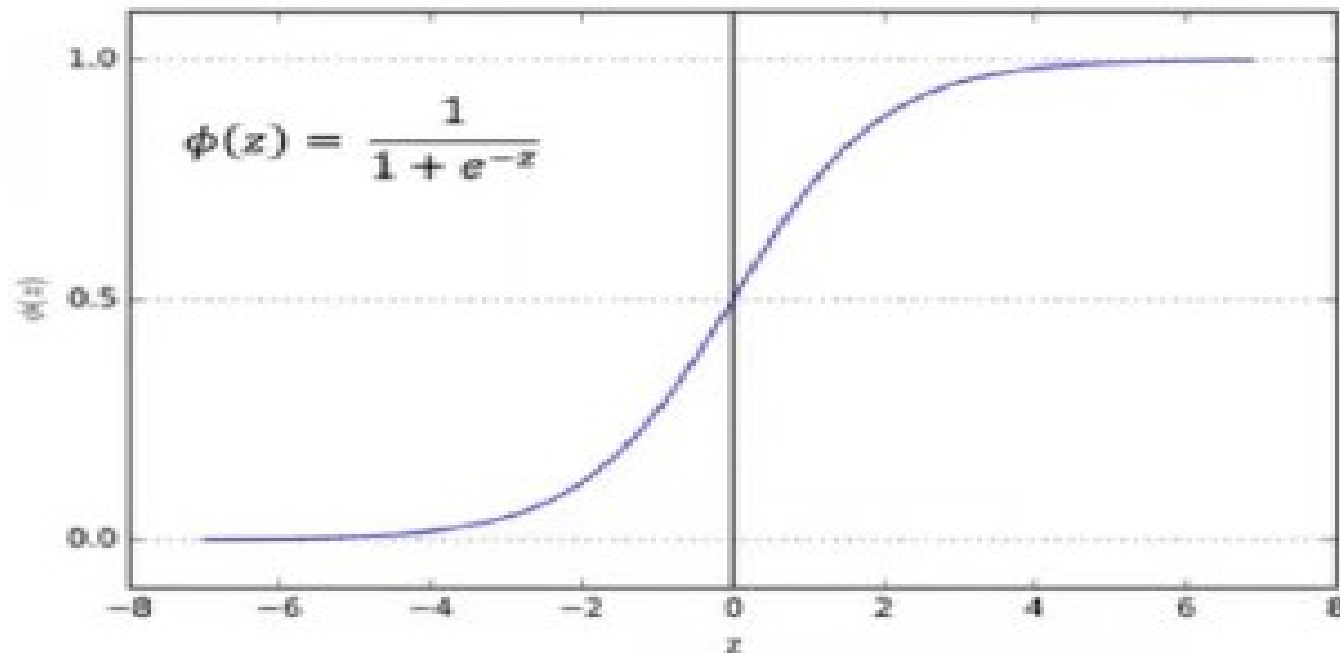


Table 1: Out-of-sample (2018-01-04 : 2019-01-28)

	Primary Model	Meta Model
Annual return		
Cumulative returns	19.7%	39.6%
Annual volatility	95.0%	56.7%
Sharpe ratio	0.65	0.82
Calmar ratio	0.29	0.96
Max drawdown	-61.9%	-36.8%
Daily value at risk	-11.7%	-7.0%

메타 레이블

□ 메타 레이블 트레이딩 예제



Example: a well calibrated (binary) classifier should classify the samples such that a `predict_proba` value of 0.8 should be approximately a 80% frequency of the positive class.

포지션 크기 결정

□ 켈리의 법칙

1	Chapter 1	1
2		2
3	THE KELLY CRITERION IN BLACKJACK, SPORTS BETTING,	3
4	AND THE STOCK MARKET ¹	4
5		5
6	EDUARDO TORRES	6
7	Edward O. Thorp and Associates, Newport Beach, CA 92660, USA	7
8		8
9	Contents	9
10		10
11	Abstract	11
12	Keywords	12
13	1. Introduction	13
14	2. Odds betting	14
15	3. Optimal growth: Kelly criterion formulas for practitioners	15
16	3.1. The probability of reaching a fixed goal in a fixed number of trials	16
17	3.2. The probability of not being selected in a fraction α of the total bankroll	17
18	3.3. The probability of being at or above a specified value at the end of a specified number of trials	18
19	3.4. Continuous approximation of expected time to reach a goal	19
20	3.5. Comparing fixed fraction strategies: the probability that one strategy beats another after n trials	20
21	4. The long run: when will the Kelly strategy "dominate"?	21
22	5. Blackjack	22
23	6. Sports betting	23
24	7. Wall street: the biggest game	24
25	7.1. Continuous approximation	25
26	7.2. The suboptimal model	26
27	7.3. The case for "fractional Kelly"	27
28	7.4. A remarkable formula	28
29	8. A word really	29
30	8.1. The constants	30
31	8.2. The strategy and results	31
32	8.3. The conclusions and the result	32
33	8.4. The theory for a portfolio of securities	33
34		34
35		35
36	¹ Paper presented at The 19th International Conference on Gambling and Risk Taking, Montreal, June 1997;	36
37	published in: <i>Fluctuating the Edge: Mathematical Analysis of Casino Games</i> , edited by G. Vassios, S.A. Gal-	37
38	ambos, W.B. Rindge, 2004. Copyrights dated April 26, 2005.	38
39		39
40	Headline of <i>Asset and Liability Management</i> , Volume 1	40
41	Edited by S. G. Dimitrakopoulou & R. Dimitrakopoulou	41
42	Copyright © 2005 Elsevier. All rights reserved.	42
43	0926-6410(2005)00006-2	43

$$f^* = \frac{\beta p - q}{\beta} \quad f^* = p - q$$

$$f^* = 2p - 1$$

Where:

- f^* = optimal bet size
- Beta = odds (win amount / lose amount)
- p = probability of success
- q = probability of failure ($1-p$)

The Investment Opportunities

Win Probability	Odds	Prob. of Selection in Simulation	Kelly Bets
0.570	1-1	0.1	0.140
0.380	2-1	0.3	0.070
0.285	3-1	0.3	0.047
0.228	4-1	0.2	0.035
0.190	5-1	0.1	0.028

[How Does the Fortune's Formula Kelly Capital Growth Model Perform?](#)

Additional papers:

- [A New Interpretation of Information Rate](#)
- [Understanding the Kelly Capital Growth Investment Strategy](#)
- [How Does the Fortune's Formula Kelly Capital Growth Model Perform?](#)
- [A Response to Professor Paul A. Samuelson's Objections to Kelly Capital Growth Investing](#)
- [Good and bad properties of the Kelly criterion](#)

Investment setting:

$$f^* = \frac{p}{\text{loss}\%} - \frac{q}{\text{profit}\%}$$

메타 레이블 구현

□ `add_vertical_Barrier(t_events, close, num_days, num_hours, num_minutes, num_seconds)`

```
add_vertical_barrier(t_events, close, num_days=0, num_hours=0, num_minutes=0, num_seconds=0)  
\[source\]
```

Advances in Financial Machine Learning, Snippet 3.4 page 49.

Adding a Vertical Barrier

For each index in `t_events`, it finds the timestamp of the next price bar at or immediately after a number of days `num_days`. This vertical barrier can be passed as an `optional` argument `t1` in `get_events`.

This function creates a series that has all the timestamps of when the vertical barrier would be reached.

- Parameters:
- `t_events` – (pd.Series) Series of events (symmetric CUSUM filter)
 - `close` – (pd.Series) Close prices
 - `num_days` – (int) Number of days to add for vertical barrier
 - `num_hours` – (int) Number of hours to add for vertical barrier
 - `num_minutes` – (int) Number of minutes to add for vertical barrier
 - `num_seconds` – (int) Number of seconds to add for vertical barrier

Returns: (pd.Series) Timestamps of vertical barriers

```
get_events(close, t_events, pt_sl, target, min_ret, num_threads, vertical_barrier_times=False,  
side_prediction=None, verbose=True) \[source\]
```

Advances in Financial Machine Learning, Snippet 3.6 page 50.

메타 레이블 구현

□ `get_events(close, t_events, pt, target, min_ret, return_threads, vertical_barrier_times, side_prediction)`

```
get_events(close, t_events, pt_sl, target, min_ret, num_threads, vertical_barrier_times=False,
side_prediction=None, verbose=True) \[source\] 
```

Advances in Financial Machine Learning, Snippet 3.6 page 50.

Getting the Time of the First Touch, with Meta Labels

This function is orchestrator to meta-label the data, in conjunction with the Triple Barrier Method.

Parameters:

- `close` – (pd.Series) Close prices
- `t_events` – (pd.Series) of `t_events`. These are timestamps that will seed every triple barrier. These are the timestamps selected by the sampling procedures discussed in Chapter 2, Section 2.5. Eg: CUSUM Filter
- `pt_sl` – (2 element array) Element 0, indicates the profit taking level; Element 1 is stop loss level. A non-negative float that sets the width of the two barriers. A 0 value means that the respective horizontal barrier (profit taking and/or stop loss) will be disabled.
- `target` – (pd.Series) of values that are used (in conjunction with `pt_sl`) to determine the width of the barrier. In this program this is daily volatility series.
- `min_ret` – (float) The minimum target return required for running a triple barrier search.
- `num_threads` – (int) The number of threads concurrently used by the function.
- `vertical_barrier_times` – (pd.Series) A pandas series with the timestamps of the vertical barriers. We pass a False when we want to disable vertical barriers.
- `side_prediction` – (pd.Series) Side of the bet (long/short) as decided by the

메타 레이블 구현

□ `get_bins(triple_barrier_events, close)`

```
get_bins(triple_barrier_events, close) \[source\]
```

Advances in Financial Machine Learning, Snippet 3.7, page 51.

Labeling for Side & Size with Meta Labels

Compute event's outcome (including side information, if provided). `events` is a DataFrame where:

Now the possible values for labels in `out['bin']` are `[0,1]`, as opposed to whether to take the bet or pass, a purely binary prediction. When the predicted label the previous feasible values `[-1,0,1]`. The ML algorithm will be trained to decide is 1, we can use the probability of this secondary prediction to derive the size of the bet, where the side (sign) of the position has been set by the primary model.

Parameters:

- `triple_barrier_events` - (pd.DataFrame) - `events.index` is event's starttime - `events['t1']` is event's endtime - `events['trgt']` is event's target - `events['side']` (optional) implies the algo's position side Case 1: ('side' not in events): bin in `(-1,1)` <-label by price action Case 2: ('side' in events): bin in `(0,1)` <-label by pnl (meta-labeling)